CardioVision AI

Multi-Vessel Risk Assessment & Clinical Vector Evaluation

RESEARCH PROTOTYPE EVALUATION REPORT — NOT FOR CLINICAL DIAGNOSIS

Evaluation Date: Sat, Oct 10, 2026, 09:19 AM

Patient ID: CV-2026-6792

Architecture: 57-Feature Four-Model Architecture

PATIENT NAME	PATIENT ID	AGE / BIOLOGICAL SEX	HEIGHT / WEIGHT / BMI
Alex Morgan (Demo Patient)	CV-2026-6792	**58 years / Male**	**170 cm / 72 kg (24.9 kg/m²)**

1. MODEL-ESTIMATED CORONARY OUTCOME PROBABILITIES

OVERALL CAD	LAD ARTERY	LCX ARTERY	RCA ARTERY
91.7%	**79.0%**	**65.0%**	**51.7%**
CAD	Stenotic	Stenotic	Stenotic

Model Estimation Notice: These are model-estimated probabilities, not clinically calibrated risk estimates or measured coronary artery narrowing percentages.

2. CLINICAL INPUT MEASUREMENTS SUMMARY

Selected patient measurements from current patient session. Machine learning prediction models evaluate the full standardized 57-feature input vector.

PARAMETER	PATIENT VALUE	GENERAL ADULT REFERENCE (INFORMATIONAL)	INFORMATIONAL STATUS
Blood Pressure (BP, Systolic)	130.0 mmHg	< 120 mmHg (resting systolic)	Above standard systolic reference (≥ 130 mmHg, Informational)
Resting Pulse Rate (PR)	70.0 bpm	60 – 100 bpm	Within standard resting range
Left Ventricular EF (EF-TTE)	50.0%	≥ 50%	LVEF ≥ 50% — interpret in clinical context
Fasting Blood Sugar (FBS)	98.0 mg/dL	70 – 99 mg/dL	Within standard fasting range
Serum Creatinine (CR)	1.0 mg/dL	0.7 – 1.3 mg/dL	Within displayed adult reference range (Informational)
LDL Cholesterol	100.0 mg/dL	< 100 mg/dL	Above standard reference (≥ 100 mg/dL, Informational)
HDL Cholesterol	39.0 mg/dL	> 40 mg/dL (M) / > 50 mg/dL (F)	Below standard reference (Informational)
Triglycerides (TG)	122.0 mg/dL	< 150 mg/dL	Within standard reference
Evaluated Model Features	**57 Model Inputs**	Standardized 7-domain feature vector	34 numeric and 23 one-hot encoded inputs

3. MODEL VALIDATION PERFORMANCE & RELIABILITY

Retrospective model evaluation metrics recorded during development using held-out test data. Not independently verified prospective clinical trial performance.

MODEL	TARGET VASCULAR OUTCOME	ACCURACY	F1 SCORE	ROC-AUC
CAD Model (`cad_model.pkl`)	Overall Coronary Artery Disease	**85.2%**	89.7%	**0.917**
LAD Model (`lad_model.pkl`)	Left Anterior Descending Stenosis	**65.6%**	67.7%	**0.756**
LCX Model (`lcx_model.pkl`)	Left Circumflex Stenosis	**57.4%**	53.6%	**0.584**
RCA Model (`rca_model.pkl`)	Right Coronary Artery Stenosis	**60.7%**	50.0%	**0.652**

Model Reliability Notice: LCX and RCA models have limited predictive discrimination in current validation results (ROC-AUC 0.584 and 0.652). Predictions should be interpreted cautiously and must not guide clinical care independently.

4. EDUCATIONAL GUIDANCE & CLINICAL CONSIDERATIONS

Scope of Evaluation & Clinical Considerations:

- Predictions represent statistical model estimates derived from patient input vectors and are not clinical diagnoses.
- Results require interpretation by a qualified physician in full clinical context.
- Actual coronary anatomy and luminal patency cannot be confirmed from machine learning probabilities alone.
- Diagnostic testing and management decisions depend on patient symptoms, physical examination, comprehensive history, and professional medical assessment.
- This application is a research prototype and is not intended for independent clinical decision-making or therapy initiation.

RESEARCH PROTOTYPE — NOT A MEDICAL DEVICE:
For educational and research demonstration purposes only. Predictions must not independently guide clinical decisions. This application is not certified for diagnostic use or treatment planning. Model probabilities and classifications do not substitute for formal diagnostic coronary imaging, physician examination, or qualified medical judgment.